

The formation of primordial black holes due to resonant processes during cosmic inflation

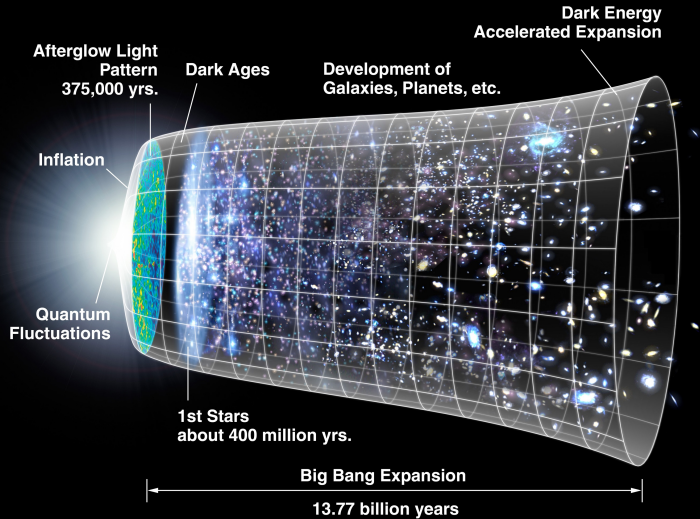
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Motivation & big picture

- Inflation: early accelerated expansion; solves horizon/flatness problems via quasi-exponential growth.
- Quantum fluctuations $\rightarrow$ curvature/density perturbations $\rightarrow$ structure formation.
- Rare large perturbations on small scales can collapse after horizon re-entry $\rightarrow$ primordial black holes (PBHs).
- Goal of this project: build numerical tools + study inflaton dynamics (first in a simplified limit), as a step toward resonance-driven enhancement relevant for PBHs.



Background: FLRW + inflation conditions

Assume homogeneity/isotropy $\Rightarrow$ FLRW metric and perfect-fluid stress tensor.

- Friedmann equations:

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}, \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p).$$

- Accelerated expansion requires $\ddot{a} > 0 \iff \rho + 3p < 0$.
- Model inflation with a scalar inflaton ϕ and potential $V(\phi)$.

Inflaton dynamics and slow-roll parameters

Scalar-field equation in an expanding background:

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0.$$

Useful parameters:

$$\epsilon \equiv -\frac{\dot{H}}{H^2}, \quad \epsilon = \frac{\dot{\phi}^2}{2M_{\text{Pl}}^2 H^2}, \quad \eta \equiv \frac{\dot{\epsilon}}{H\epsilon}.$$

- Inflation: $\epsilon < 1$; end of inflation: $\epsilon = 1$.
- Slow-roll: $\dot{\phi}^2 \ll V(\phi)$ and $\epsilon, \eta \ll 1$.
- Kination (opposite regime): kinetic energy dominates, $\rho_\phi \propto a^{-6}$.

Reheating vs preheating

After inflation, the inflaton must transfer energy to matter degrees of freedom.

- **Reheating (perturbative)**: inflaton decays with rate Γ ; radiation bath forms, rough end when $H \sim \Gamma$.
- **Preheating (non-perturbative)**: early stage with resonant instabilities and rapid particle production.
- A typical example: coupling to a scalar χ gives time-dependent effective frequency $\omega_k(t)$ for modes χ_k .

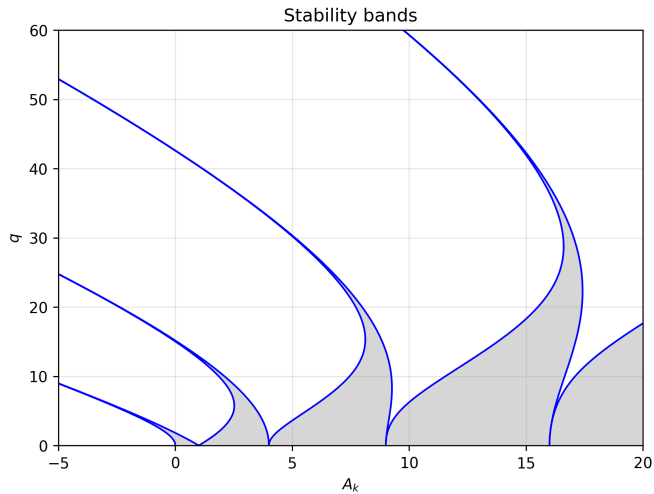
Parametric resonance and Mathieu equation

In a simplified preheating setup (neglect expansion over short times), mode equation becomes Mathieu:

$$\chi_k'' + (A_k - 2q \cos 2z) \chi_k = 0, \quad A_k = \frac{k^2}{m^2} + 2q, \quad q = \frac{g^2 \Phi^2}{4m^2}.$$

- Floquet theory $\Rightarrow$ instability bands where χ_k grows exponentially.
- Particle production tracked via comoving occupation number n_k (computed from mode functions and ω_k).

Stability curves



PBHs from metric perturbations: Newtonian gauge picture

Perturbed FLRW in Newtonian gauge:

$$ds^2 = -(1 + 2\Phi) dt^2 + a(t)^2(1 - 2\Phi)\delta_{ij}dx^i dx^j.$$

- Field perturbations $\delta\phi_k, \delta\chi_k$ couple to metric perturbation Φ_k .
- From solutions one constructs curvature perturbation ζ_k and its power spectrum:

$$P_\zeta(k) = \frac{k^3}{2\pi^2} |\zeta_k|^2 \quad \text{with} \quad \zeta_k = \Phi_k + 2H \frac{\dot{\Phi}_k + H\Phi_k \left(1 + \frac{1}{3} \frac{k^2}{a^2 H^2}\right)}{\dot{\phi}^2 + \dot{\chi}^2}.$$

- PBH idea: if small-scale $P_\zeta(k)$ is enhanced, overdense regions can collapse upon horizon re-entry (radiation era).

Numerics: RK4 strategy

We solve ODE systems by rewriting second-order equations into first-order vector form:

$$x'' = F(x, x', t) \quad \Rightarrow \quad \begin{cases} x' = v, \\ v' = F(x, v, t). \end{cases}$$

RK4 update (conceptually): use four slope evaluations per step and a weighted average to advance the state.

- Initialize $\dot{\phi}_0$ from slow-roll estimate.
- Integrate until end of inflation determined by $\epsilon = 1$.
- Record N_{end} and diagnostics (e.g., particle-production indicator).

Model used in Results: potential + extended field content

Lagrangian includes inflaton ϕ and complex scalar ψ with self-interaction:

$$\mathcal{L} = \frac{1}{2} \nabla_\mu \phi \nabla^\mu \phi + \nabla_\mu \psi^* \nabla^\mu \psi - V(\phi) - \frac{1}{2} m_\psi^2(\phi) |\psi|^2 - \frac{\lambda}{4} |\psi|^4 - \frac{\lambda}{4} f^4,$$

with

$$m_\psi^2(\phi) = g^2 (\phi - \phi_{\text{SBP}})^2 - \lambda f^2.$$

Inflationary potential studied (exponential-to-exponential form):

$$V(\phi) = V_0 \exp \left[-\kappa \exp \left(\frac{\phi}{b M_{\text{Pl}}} \right) \right].$$

Shape of the potential

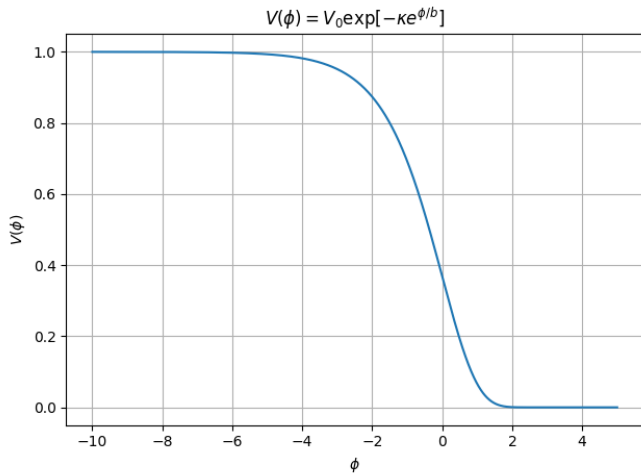
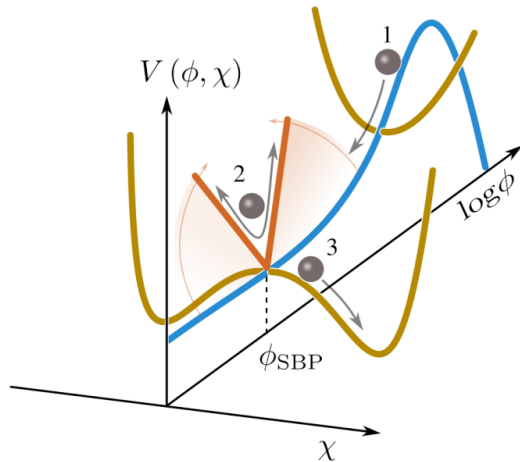


Illustration of the potential



Derived equations

$$\ddot{\phi} + 3H\dot{\phi} - \frac{\kappa V_0}{bM_{\text{Pl}}} \exp\left[\frac{\phi}{bM_{\text{Pl}}} - \kappa e^{\frac{\phi}{bM_{\text{Pl}}}}\right] + g^2\langle\chi^2\rangle(\phi - \phi_{\text{SBP}}) = 0$$

$$\ddot{\tilde{\chi}}_{\mathbf{k}} + 3H\dot{\tilde{\chi}}_{\mathbf{k}} + \left[\frac{k^2}{a^2} + g^2(\phi - \phi_{\text{SBP}})^2 - \lambda f^2 + 3\lambda\langle\chi^2\rangle + \langle\dot{\theta}^2\rangle + \frac{1}{a^2}\langle(\nabla\theta)^2\rangle\right]\tilde{\chi}_{\mathbf{k}} = 0$$

$$\langle\chi^2\rangle\ddot{\tilde{\theta}}_{\mathbf{k}} + (2\langle\chi\dot{\chi}\rangle + 3H\langle\chi^2\rangle)\dot{\tilde{\theta}}_{\mathbf{k}} + \frac{k^2}{a^2}\langle\chi^2\rangle\tilde{\theta}_{\mathbf{k}} = 0$$

where the averages are

$$\langle\chi^2\rangle = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} [|\tilde{\chi}_{\mathbf{p}}(t)|^2 + 2\text{PI-corrections}]$$

$$\langle\dot{\theta}^2\rangle = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} [|\dot{\tilde{\theta}}_{\mathbf{p}}(t)|^2 + 2\text{PI-corrections}]$$

Analytic checks (slow-roll and kination limits)

Slow-roll in $N = \ln a$ gives a closed-form reference solution (used to validate numerics):

$$\phi(N) = -b \ln \left(\frac{\kappa}{b^2} (N_0 - N) \right).$$

Kination attractor (kinetic dominated):

$$\phi(N) = \phi_* \pm \sqrt{6(N - N_*)}.$$

- These limits provide sanity checks: numerical $\phi(N)$ should match slow-roll early, and approach kination-like behavior when appropriate.

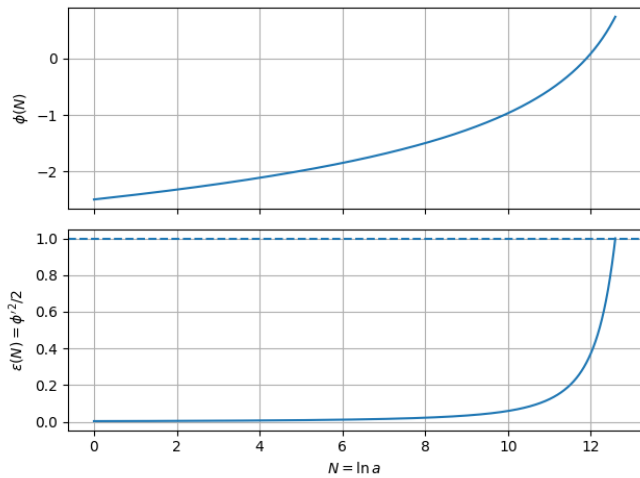
Numerical inflaton evolution and end of inflation

In the simplified **inflaton-only** approximation ($\langle\chi\rangle = \langle\theta\rangle = 0$), RK4 integration yields:

- Rapid growth of $\phi(N)$ in the chosen potential.
- End of inflation at approximately:

$$N_{\text{end}} \approx 12.59 \quad (\epsilon = 1).$$

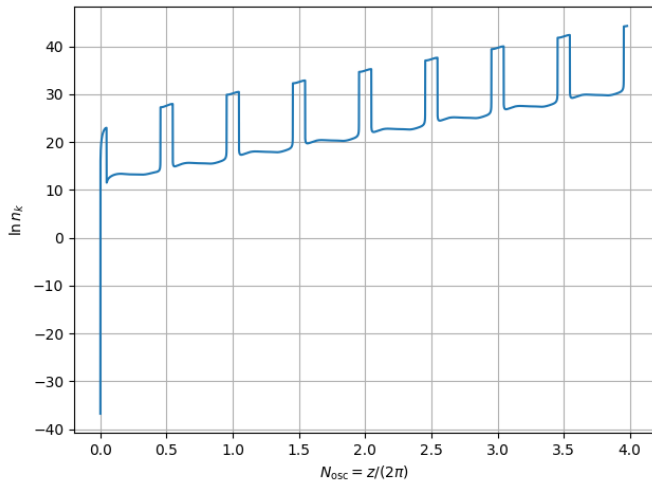
- Post-inflation dynamics are unrealistic without energy transfer/backreaction (only Hubble friction present).



Particle-production indicator and interpretation

Comoving occupation-number indicator shows step-like growth:

- In comoving variables, $\ln n_k$ can exhibit jumps and temporary drops (coordinate/definition artifact).
- In a more complete treatment (including coupled χ, θ + backreaction), effective friction and energy transfer should modify both $\phi(N)$ and particle production.



Takeaways & next steps toward PBHs

- Built RK4-based framework and validated inflaton evolution against analytic limits.
- Found $N_{\text{end}} \approx 12.59$ for the inflaton-only evolution in this potential.
- Next steps:
 - Couple in χ, θ dynamics and compute averages $\langle \chi^2 \rangle$,
 - Track resonance-driven amplification and propagate into metric perturbations Φ_k and $P_\zeta(k)$.
 - Identify parameter regions where small-scale $P_\zeta(k)$ is enhanced enough to support PBH formation.

Key references

-  L. Kofman, A. Linde, A. A. Starobinsky, *Phys. Rev. D* 56 (1997).
-  Y. S. Furuta et al., *Primordial Black Holes as Dark Matter and the Tachyonic Trap During Inflation* (2025).